NATIONAL UNIVERSITY OF SINGAPORE

Activity-Based Risk Assessment Form

Name of Department	**Department of Biological Sciences**	**Location of Lab**	**Block S3 Level 4**
Name of Laboratory	**Reef Ecology Laboratory**	**Name of PI**	**Huang Danwei**
Name of Researcher	**Lutfi Afiq Rosli**	**Name of Activity/Experiment**	**Overseas field work**

No	Description/Details of Steps in Activity	Hazards	Possible Accident / Ill Health & Persons-at-Risk	Existing Risk Control (Mitigation)	Severity	Likelihood (Probability)	Risk Level	Additional Risk Control	Person Responsible	By (Date)
1	Loading, unloading, embarking and disembarking vessel	Falling on land while embarking or disembarking vessel	Bruising to passenger	(a) Make sure boat is tied down and immobile before boarding (b) Ensure boat carries life jackets and First Aid kit. (c) For shore entry on small boats, ensure proper balance before hopping on board or use a ladder from waist-deep water if available.	1	1	**1**			
		Falling into water while embarking or disembarking vessel	Drowning of researcher		2	1	**2**			
2	Adverse weather	Working under the hot sun	Heat injuries, dehydration, sunburn on passenger	(a) Monitor weather conditions before going to field (b) Ensure sufficient water intake (c) Ensure adequate rest (d) Avoid outer deck as long as possible during the trip (e) Apply sunblock, wear hat and sunglasses if working on deck (f) Ensure researchers are trained in First Aid for handling heat injuries	1	1	**1**			
			Heat injuries could lead to heat exhaustion and stroke to passenger		3	1	**3**			
		Lightning risk	Lightning strike; electrocution leading to fatality of passenger	(a) Plan trip through reviewing weather forecast few days ahead (b) Monitor weather conditions before going to field (c) Seek shelter on land during inclement weather (d) Do not board with forecast of inclement weather (e) Ensure lighting protection system on the vessel is in working order	3	1	**3**			

3	Rough seas	Seasickness experienced by passenger in rough seas	Discomfort to passenger	(a) Avoid outer deck as long as possible during the trip (b) Ensure good ventilation in cabin (c) Encourage passengers to bring aboard own seasickness medication	1	1	**1**			
		Man overboard in rough seas	Man overboard during rough sea conditions leading to drowning of passenger	(a) Monitor weather conditions before going to field (b) Hold on to handrails (c) Ensure boat carries life jackets (d) Seek shelter on land during inclement weather (e) Do not board with forecast of inclement weather (f) Ensure vessel is surrounded by guard rails (g) Ensure passengers are proficient swimmers (h) Establish emergency response procedure for man overboard situations	3	1	**3**			
4	Prolonged dive session	Shallow breathing during prolonged diving leading to diver running out of air	Hypercapnia leading to panic, unconsciousness and eventually fatality to diver; drowning leading to fatality to diver	(a) Be disciplined, do not overload work within one dive (b) Minimum 50 bar of air should be on reserve and divers should surface (c) Buddy to supply air to distressed diver in an emergency, secure the spare regulator by holding on to the hose (d) Perform pre-dive briefing and safety check before diving: emphasising dive duration, tasks, procedures and safety measures	3	1	**3**			

		Carbon dioxide build up and lack of drinking water while diving	Headache; dehydration to the diver	(a) Take full even breaths with the regulator and avoid 'skip breathing' (b) Hydrate adequately before and in between dives (c) Stop dive if headache is severe and persists after surfacing and resting (d) No more than three dives per day, and return to boat/shore on any sign of tiredness	1	2	**2**			
5	Entering water while carrying heavy tools and load	Rapid descent of diver due to handling of heavy equipment	Sinus squeeze, burst or stretched ear drum, and discomfort to diver due to rapid descent	(a) Do not carry excessive load during dive (b) Inflate BCD and regulate buoyancy for gradual descent (c) Lower heavy equipment to underwater site from vessel via ropes (d) Perform checkout dive before handling heavy equipment (e) Ensure ears can be equalised properly before dive (f) No diving with common cold or allergies that cause nasal congestion (g) No more than three dives per day, and return to boat/shore on any sign of tiredness.	2	1	**2**			
6	Ascending to surface at the end of dive	Rapid ascent of diver	Blocked sinus duct and middle ear eustachian tube leading to middle ear barotrauma to diver	Ensure ears can be equalised properly before dive	2	1	**2**			
			Failing to maintain open airway during ascent leading to pulmonary barotrauma to diver	(a) No holding of breath while ascending (b) Ascend at speed of 1 m per 4 s or follow the last smallest bubble (c) No diving with blocked nose or ears	3	1	**3**			

			Accumulation of nitrogen loads leading to decompression sickness	(a) Good hydration, adequate rest and no strenuous activity (b) No flying for 24 hrs post-dive (c) Diver to monitor for symptoms for 24 h post dive (d) Emergency oxygen first aid available	2	1	**2**			
		Colliding of diver with other vessels in area of heavy boat traffic	Injuries to diver from collision with vessel	(a) Deploy dive flag on dive vessel during dive (b) Deploy surface safety marker prior to ascent in areas of heavy boat traffic (c) Ascend at speed of 1 m per 4 s or follow the last smallest bubble (d) Raise arm during ascent to ensure no overhead obstructions	3	1	**3**			
7	Diving amongst algae, seaweed or abandoned fishing nets	Entanglement by fishing nets and lines, or thick algae and seaweed	Time taken for disentanglement could lead to out of air situation, leading to panic, unconsciousness and fatality	(a) Carry a dive knife to cut entanglement (b) Situational and dive site awareness (c) Perform checkout dive before work dive (d) Carry enough reserve air to allow reasonable amount of time to handle emergencies (e) Buddy to supply air to distressed diver in an emergency, secure the spare regulator by holding on to the hose	3	1	**3**			
8	Diving amongst marine animals	Hard corals, sharp rock edges and dangerous marine animals (e.g. hydroids, jellyfish, venomous fish)	Cuts, infected wounds to diver	(a) Wear wet suit, booties, gloves (b) No contact with reef unless contact point positively identified (c) If bitten, try to note the marine organism that attacked to assist medical treatment	1	1	**1**			
			Cuts, poisoning and pain from poisonous stings and bites to diver		2	1	**2**			

9	Diving in poor visibility (<0.5 m)	Minor wounds from hitting reef	Cuts, infected wounds to diver	(a) Don full-body wetsuit and gloves (b) Carry whistle, tank banger and surface safety marker	1	2	**2**			
10	Swimming at the surface to point of descent while diving	Prolonged surface swimming	Overexertion; hyperthermia	(a) Vessel deploy divers only if site is within 10 min swim given existing dive conditions (b) Vessel collect divers if still at surface after 10 min	1	1	**1**			
11	Diving in unfamiliar reef/ water using rental gears.	Strong currents (>1 knots), lost of orientation, encounter of dangerous marine animals, colliding of divers with other vessels, lightning and rough seas risk, equipment malfunction, boat malfunction, dirty compressed air	Cuts, poisoning and pain from poisonous stings and bites to diver, overexertion, hyperthermia, injuries from colliding with vessel, infected wounds, out of air situation, drowning leading to fatality of diver	(a) Ensure proper communication with dive operators (b) Present viable dive insurance to dive operators (c) Learn emergency assistant plan (EAP) of the dive operators (d) Test equipment before diving. Do not dive with faulty equipment. (e) Ensure that the boat has at least two working communication device (radio transmitter, mobile phone, satellite phone, etc). (f) Ensure boat carries life jackets and First Aid kit. (g) Plan dives together with dive operators at least a few hours before dive operations. (h) Conduct dive briefings with dive operators before each dive. (i) Adhere to advice from dive operators at all times. (j) Adhere to safe diving practices.	3	1	**3**			
12	Diving in open waters	Strong currents (>1 knots)	Overexertion; hyperthermia	(a) Vessel to move down current and deploy safety marker buoy (b) Diver carry whistle, tank banger and surface safety marker (c) Estimate current speed during initial phase of dive to determine if dive needs to be terminated	1	1	**1**			

13	Diving on deep patch reefs	Lost of orientatation and bearing.	Out of air, panic, buddy lost, overexertion.	(a) Conduct dive briefings prior to entry. (b) Carry underwater compass to ensure correct bearing at all times. (c) Stay calm at all times (d) Monitor air supply regularly (e) When sharing air with alternate air source, diver to hold on to second stage while ascending	2	2	**4**			
14	Use of pressurised air cylinder for SCUBA diving	Pressurised cylinder explosion; 'rocketing' uncontrolled runaway cylinder	Injuries to diver from explosion or runaway cylinder	(a) Check date of servicing, testing and charging of the diving gas cylinder before dive (b) Ensure dive does not start with a gas cylinder with pressure >>200 bar (c) Buddy to supply air to distressed diver in an emergency, secure the spare regulator by holding on to the hose (d) Maintain all dive equipment properly, with annual serving of regulators	2	1	**2**			
		Burst demand valve hose and damaged dive regulator	Damaged regulator underwater would lead to cut off of air supply and out of air		2	1	**2**			
15	Collection of water sample	Slipping and falling due to slippery surface	Bruising and abrasions to researcher	(a) Wear covered shoes with good grip (b) Move carefully and be aware of surroundings (c) Bring stocked First Aid kit (d) Bring emergency contacts for boats and evacuation, and charged mobile phone	1	1	**1**			
			Sprains or fractures to researcher		2	1	**2**			
16	Sampling at sites with strong flowing water	Falling into open water, man overboard situation and/or swept away by currents	Drowning, leading to fatality of researcher	(a) Ensure that researcher is accompanied (b) Employ harness linked to guardrails (c) Ensure researcher is able to swim (d) Bring emergency contacts for boats and evacuation, and charged mobile phone	3	1	**3**			

17	Sampling at sites with no shelter	Working under the hot sun	Heat injuries, dehydration, sunburn on researcher	(a) Monitor weather conditions before going to field (b) Ensure sufficient water intake (c) Ensure adequate rest (d) Apply sunblock, wear hat and sunglasses if working on deck (e) Ensure researchers are trained in First Aid for handling heat injuries	1	1	**1**			
			Heat injuries could lead to heat exhaustion and stroke to researcher		3	1	**3**			
		Lightning risk	Lightning strike; electrocution leading to fatality of researcher	(a) Plan trip through reviewing weather forecast few days ahead (b) Monitor weather conditions before going to field (c) Seek shelter during inclement weather (d) Do not proceed with collection with forecast of inclement weather	3	1	**3**			
18	Movement in field sites	Stepping on or agitation of hidden venomous and dangerous animals (e.g. stonefishes, stingrays, sea snakes)	Venomous stings, bites or puncture wounds to researcher	(a) Move slowly and cautiously (b) Highlight and note potentially dangerous animals, surfaces and hazards (c) Wear thick sole booties (d) Bring stocked First Aid kit (e) Bring emergency contacts for boats and evacuation, and charged mobile phone	2	1	**2**			
		Stepping on hazardous debris (e.g. fishing line, sharp glass)	Cuts and puncture wounds to researcher		1	1	**1**			
		Stepping on slippery or uneven surfaces and falling	Sprains or fractures to researcher		2	1	**2**			
19	Movement in sites with no shelter	Prolonged exposure to sun	Heat injuries, dehydration, sunburn on researcher	(a) Ensure researcher is hydrated (b) Ensure researcher is protected against the sun with sunblock or covered clothing (c) Ensure researchers are trained in First Aid for handling heat injuries	1	1	**1**			
			Heat injuries could lead to heat exhaustion and stroke		3	1	**3**			

		Inclement weather and lightning strikes in open area at sea	Lightning strikes on researcher leading to death of researcher	(a) Monitor weather conditions prior to field work (b) Cancel or postpone field work in inclement weather conditions (c) Stop activity and seek shelter at jetty or huts if inclement weather occurs during field work	3	1	**3**			
20	Movement in intertidal area near open water	Falling into open water or swept away by currents	Drowning leading to fatality of researcher	(a) Ensure researcher is able to swim (b) Ensure researcher does not wade deeper than waist-deep water (c) Stop activity when tides are rising fast	3	1	**3**			
21	Collection of intertidal organisms	Usage of sharp tools and contact with sharp surfaces and dangerous organisms	Cuts and puncture wounds to researcher	(a) Ensure proper handling of tools (b) Ensure proper handling of organisms	1	1	**1**			
22	Preservation and transport of intertidal and subtidal organisms	Exposure to ethanol	Liquid, vapours, fumes or mist may cause irritation. Direct contact may cause redness and pain. Large doses ingested may lead to alcohol poisoning. Prolonged or frequent contact may result in liver injury. Intoxicating if continuously inhaled for a long period of time.	(a) Wear latex or nitrile gloves, safety glasses, and lab coat (b) Keep chemicals away from exposed skin and eyes (c) Do not work with more than 2.5 litres of these chemicals (d) Tighten specimen container properly and pack containers in an enclosed box. Label all specimens and boxes accordingly.	2	1	**2**			
		Exposure to magnesium chloride	Slight irritation upon skin or eye contact, or upon inhalation		1	1	**1**			
		Ethanol is volatile and flammable	Can cause skin burns or fire	(a) Wear latex or nitrile gloves, and lab coat (b) Store in approved flammable liquid storage containers, away from incompatible materials (c) Only to be used in a cool, dry well-ventilated area away from sparks, flames and other sources of ignition (d) Eliminate all sources of static	3	1	**3**			

		Exposure to dimethyl sulphoxide (DMSO)	Irritation on skin, irratation on lungs if inhaled excessively	(c) Eliminate all sources of static electricity	1	1	1			

Conducted By

Name: Lutfi Afiq Bin Rosli

Signature:

Approval date: 3/12//2018

Approved By

Name: Huang Danwei

Signature:

Approval date: 03-12-2018

Next Revision date (Maximum 3 years): 01-07-2021

		Likelihood		
		Likely	Possibly	Unlikely
Severity	Low	3	2	1
	Med	6	4	2
	High	9	6	3

Risk = Likelihood x Severity

RISK	DECISION PROCESS
< 3	**RISK ACCEPTABLE**
3, 4	**CONSIDER ADDITIONAL RISK CONTROL**
> 4	**ADDITIONAL RISK CONTROL REQUIRED**

Likelihood

1	**Unlikely**	Not likely to occur (has not occurred in the PI's Lab or similar Lab setup.)
2	**Possible**	Possible or known to occur (has occurred in the PI's Lab or Similar Lab setup.)
3	**Very Likely**	Common or repeating occurrence (has occurred repetitively in the PI's Lab or similar Lab setup.)

Severity

1	**Low**	(e.g. No injury, injury or ill-health requiring first aid treatment only - includes minor cuts and bruises, irritation, ill-health with temporary discomfort)
2	**Medium**	(e.g. Injury requiring medical treatment or ill-health leading to disability – includes lacerations, burns, sprains, minor fractures, dermatitis, deafness, work-related upper limb disorders)
3	**High**	(e.g. Fatal, serious injury or life-treatening occupational disease – includes amputations, major fractures, multiple injuries, occupational cancer, acute poisoning and fatal diseases)

Severity - Consider the magnitude/severity of the consequences of the Risk Factor occurring and then list this as 3 (High), 2 (Moderate) or 1 (Low).
Severity normally will not change unless there is a physical change to the equipment or process.

Likelihood - Team should rely upon their experience and consider realistic scenarios. Listed below are examples of factors that may be considered in determining the likelihood.
- Past experience / incidents
- Complexity of the activity
- Number of personnel involved in the activity (e.g. all personnel, a limited number of trained personnel, etc)
- Frequency of use or execution
- Degree of control (involvement of contractors)
- Strength/completeness of administrative controls
- Sufficiency/formality of training
- Other....